

UNIVERSITY OF BOTSWANA

DEPARTMENT OF MATHEMATICS

2013/2014 – EXAMINATIONS

COURSE NUMBER MAT111 DURATION 2 hrs DATE MAY 2014

TITLE OF PAPER INTRODUCTORY MATHEMATICS I

SUBJECT MATHEMATICS TITLE OF EXAMINATION BSc I

MORNING/AFTERNOON

INSTRUCTIONS:

- ANSWER ALL QUESTIONS IN SECTION A AND ANY TWO(2) QUESTIONS FROM SECTION B.
- ALL MARKS ARE INDICATED IN BRACKETS [].

NUMBER OF PAGES INCLUDING COVER PAGE: 5

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TO DO SO BY THE SUPERVISOR.

SECTION A

Answer All Questions: Each Question carries 10 Marks

Question 1

(a) Express the number $3.47\bar{5}$ in the form $\frac{p}{q}$ where, $p, q \in \mathbb{N}$. [5]

(b) Simplify

$$\frac{2^{1/2}}{2^{1/2} + \sqrt{3}} - \frac{3^{1/2}}{\sqrt{2} - 3^{1/2}}.$$

[5]

Question 2

(a) Solve:

$$2 \log_5 4x^2 = 4.$$

[5]

(b) Solve:

$$x^2 - x - 6 \leq 0.$$

[5]

Question 3

(a) Solve the following simultaneous equations

$$2y + x + 1 = 0$$

$$6y + 2x = 4.$$

[5]

(b) Let $f(x) = 4x - \frac{1}{x}$, $g(x) = \frac{1}{x}$. Determine the function $(f + g)(x)$ and find its domain.

[5]

Question 4

(a) Find the real number a if the equation $(a - 1)x^2 - 5x = 1$ has two equal real roots. [5]

(b) Sketch the graphs of $f(x) = |x^2 + x - 2|$ and $g(x) = |x^2 + x - 2| + 2$ on the same axes. [5]

Question 5

(a) Find the values of b such that when the function $f(x) = x^2 - 5x - 4$ is divided by $x + b$ the remainder is 2. [5]

(b) Completely factorize the function $f(x) = x^3 - 3x - 2$. [5]

Question 6

(a) Find the exact value of

$$\cos\left(\frac{22\pi}{3}\right).$$

[4]

(b) Determine the value of $\operatorname{cosec}(-\theta)$ given that $\cos\theta = \frac{2}{3}$ for $\frac{3\pi}{2} < \theta < 2\pi$.

[6]

SECTION B

Answer TWO(2) Questions OUT of FOUR(4):
Each Question carries 20 Marks

Question 7

(a) Solve:

$$2^{2x-1} = 3^{3x+1}.$$

[5]

(b) Sketch the graph of

$$f(x) = \frac{1}{(x-2)(x+3)}.$$

[7]

(c) Given that the roots of $x^2 + 3x + 1 = 0$ are α and β , find, without solving this equation, the quadratic equation whose roots are $2\alpha - \beta$ and $2\beta - \alpha$.

[8]

Question 8

(a) Determine whether the function $f(x) = x^3 - x^2$ is odd, even or neither.

[4]

(b) Find the maximal domain and range for the function

$$f(x) = \sqrt{\frac{x-2}{x+1}}.$$

[10]

(c) Show that the function $f : \mathbb{R} \longrightarrow (0, \infty)$ given by $f(x) = e^x$ is bijective and hence find $f^{-1}(x)$.

[6]

Question 9

- (a) Solve the inequality and express your answer in interval notation:

$$\frac{x^2 - 1}{x - 5} > -1.$$

[7]

- (b) Solve:

$$\sqrt{x} - \sqrt{x - 2} = 1.$$

[5]

- (c) Show that $a^2 + b^2 + c^2 + d^2 > 2(ab + cd)$ for pairwise distinct $a, b, c, d \in \mathbb{R}$.

[4]

- (d) Simplify

$$\frac{x^2 - 9}{x^2 - 2x - 15} \div \frac{x^2 - 6x + 9}{12 - 4x}.$$

[4]

Question 10

- (a) Let $z_1 = 1 + i$, $z_2 = -1 + 2i$.

- (i) Express z_1 in polar form.

[4]

- (ii) Compute $\frac{z_1}{z_2}$ and express your answer in the form $a + bi$.

[4]

- (b) Prove the following trigonometric identities.

$$(i) \frac{1 + \sec \theta}{\sin \theta + \tan \theta} = \csc \theta \quad (ii) \sin^4 \theta - \cos^4 \theta = 2 \sin^2 \theta - 1.$$

[3,3]

- (c) Solve the equation

$$2 \sin^2 \theta + \cos \theta = 1$$

in the interval $0 \leq \theta \leq \pi$.

[6]